



High Speed Vendor Feed

MX SOLA® HSVF UDP Multicast

Specifications Guide

Protocol D7

Confidential

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Document History

Version	Date	Change Description
1.0	2014-05-01	Initial document
1.1	2014-08-19	Section 6, Tick Table has been modified and re-worked. Field Name 'Stamp Time' has been changed to 'Timestamp' in messages 'C', 'CB', 'CF', 'CS', 'I', 'IB', 'IF', 'IS'. Definition of Timestamp for messages: 'C', 'CB', 'CF' and 'CS' is 'Time of the transaction' Definition of Timestamp for messages; 'I', 'IB', 'IF', and 'IS' is 'Time of the cancellation transaction'
1.2	2015-02-04	Section 3.2.19, modified Message Type FB
1.3	2015-04-02	Modification to Section 2.6, Retransmission, first note Sections 3.2.38, 3.2.39, 3.2.40, 3.2.41: Modification of Data type for Bid/Ask Size fields
1.4	2015-08-17	Modification to Section 2.6, TCP Retransmission Capability, third note Section 3.2.2, Message Type LI, remove D3 protocol, add D4 protocol Section 3.2.14, add Message Type CW Section 3.2.19, add Message Type DW Section 3.2.24, add Message Type FW Section 3.2.31, add Message Type HW Section 3.2.36, add Message Type IW Section 3.2.48, add Message Type NW Section 3.2.53, add Message Type QW Section 3.2, add Market Side to Message Type D tables Section 5.8, add Protocol Version table Section 5.9, add Frequency Codes table Section 6.1, add Swap Future tick increment Section 8.3, delete Market Feed Indicators table Section 5.10, add Market Feed Indicators table Section 5.10 add S - Swap to Market Feed Indicators table
1.5	2016-08-01	Correction to Message Type IW (Market Price Indicator) General editing
1.6	2017-07-05	HSVF Improvement: D5 Protocol Section 3.4.1 to 3.4.5: Updates to messages C/CB/CF/CS/CW

		Section 3.5.2 to 3.5.5: Updates to messages DB/DF/DW Section 3.6.2 to 3.6.5: Updates to messages FB/FF/FW Section 3.7.2 to 3.7.5: Updates to messages HB/HF/HW Section 3.8.1 to 3.8.5: Updates to messages I/IB/IF/IS/IW Section 3.9.1 to 3.9.6: Updates to messages J/JB/JF/JS/JW Section 3.10.1 to 3.10.5: Updates to message N/NB/NF/NS/NW Section 3.12.5: New message Tick Table TT Section 3.12.8: New message Timebeat Z Grouped field descriptions contained in these sections (4.1, 4.2, 5.1 to 5.10, 6) under new Section 4- Fields Description Section 4.4 (formerly 5.3): Price Indicator updated Section 4.9 (formerly 5.8): D5 added Section 5 (formerly Section 7): J/JB/NS examples updated
	2017-08-14	Section 3.4.2: Correction to CB message Section 3.4.5: Correction to CW message Section 3.7.1: Correction to H message Section 3.9.6: Correction to JW message
1.7	2017-09-15	Section 3.4.1 to 3.4.3: Modifications to C, CB, CF message Section 3.4.9: Modification to JF Section 3.10.1, 3.10.2: N, NB messages updated Section 3.12.1 to 3.12.5: E, EB, EF, ES, EW messages added Section 4.13: Tick Table updated Appendix A: Tick Table added
1.8	2017-09-26	Section 3.12.5: Message Tick Table TT updated Remove Appendix A
	2017-10-16	Section 3.4.3: CF message updated (add paragraph) Section 3.9.6: JW message updated (symbol includes Tenor, Fixed Rate, Fixed Rate Fraction Indicator) Section 3.10.1 to 3.10.5 : N, NB, NF, NS, NW messages updated (Price format, U value description) Section 4.4: Price Indicator Markers updated (Q, V description) Section 4.12: Update to table heading Section 5.2: Update to Example 2
1.9	2017-12-20	Moved Section 3.13 Instrument Schedule messages to 3.6 Added Section 3.13: Trade Correction Messages X, XB, XF, XS, XW
	2018-06-05	Section 4.4 Added Price Indicator Marker value G
	2018-07-13	Section 3.11.2 Message Type NB: Updates to length of fields Underlying Symbol and Delivery Year

1.10	2018-09-11	Section 2.2 HSVF Feed Schedule of a Typical Day updated for extended trading hours
1.11	2019-01-23	Section 3.11.2 Message Type NB: Underlying Symbol field length updated to 3 bytes and Delivery Year field length updated to 1 byte (Roll back to Version 1.9) Update to Note 1, p 64
1.12	2020-01-20	• Protocol D6 – Diffusion enhancements ○ Enlarge price to support 4 decimals ○ Strategy structure in the dictionary ○ Additional Market Feed indicators ○ Open Interest date ○ Strategy profile codes ○ Expiry date for Futures Options ○ Microsecond timestamps ○ Enlarge the underlying field of the J messages (Option Instrument Key) to support 12 characters ○ Remove Open Interest field from trades • Removed references to Swaps • Updated the Error List • Typos and other minor corrections
1.12.1	2020-03-18	• Update JF and NS message length
1.12.2	2020-06-15	• New note in Section 4.12 Tick Table, as trade prices can be more granular (up to 4 decimals) than the Tick Table • Fields and Description changes in business messages containing 'Symbol blocks' for Expiry Dates and Symbol Dates, to standardize per type of products
1.13	2021-12-31	• Protocol D7 • Extended JS message for covered trade • Provide instruments Last Trading Date Time (J,JS,JF,JB) • Provide Bonds deliverable new message (KF) • Business date broadcast (SD) • Extended Sequence Number (headers + RT) • Rename Market Feed to Market Flow
1.13.1	2022-08-18	• Strategy Code • Enlarged header timestamp • Removed Trade timestamp from C,CB,CF,CS,I,IB,IF,IS,XF - HeaderTime is now = to timestamp, and improved time precision to microseconds. • Strategy type definition
1.13.2	2022-09-15	• Corrected missing field in SD message
1.13.3	2022-10-05	• Market Flow Indicators: new Type of Underlying

		○ I: Inter Sectors
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Table of Contents

Introduction	10
Objective	10
Scope	10
Products Information	10
Technical Support	10
Trading Overview	11
Connectivity	11
Interface	11
HSVF Feed Message Sequence of a Typical Day	11
Transmission Format	12
Data Format	12
TCP Retransmission Capability	14
Intraday Failover to Disaster Recovery Site	15
Messages	15
Message Types	16
Conventions	19
TCP (Retransmission) Messages	19
LI: Login (52 Bytes)	19
LO: Logout (12 Bytes)	19
KI: Login Acknowledgement (12 Bytes)	20
KO: Logout Acknowledgement (12 Bytes)	20
RT: Retransmission Request (34 Bytes)	20
RB: Retransmission Begin (12 Bytes)	20
RE: Retransmission End (12 Bytes)	20
ER: Error Message (96 Bytes)	20
Trade Messages	21
C: Option Trade (86 Bytes)	21
CB: Future Options Trade (87 Bytes)	21
CF: Futures Trade (78 Bytes)	23
CS: Strategy Trade (98 Bytes)	23
Request for Quote (RFQ) Messages	24
D: Option Request for Quote (RFQ) (61 Bytes)	24
DB: Future Options Request for Quote (RFQ) (62 Bytes)	25
DF: Futures Request for Quote (RFQ) (53 Bytes)	26
DS: Strategy Request for Quote (RFQ) (72 Bytes)	26
Instrument Schedule Notice Messages	27
E: Instrument Schedule Notice Option (59 Bytes)	27
EB: Instrument Schedule Notice Futures Option (60 Bytes)	27

EF: Instrument Schedule Notice Future (51 Bytes)	28
ES: Instrument Schedule Notice Strategy (70 Bytes)	28
Quote Messages	29
F: Option Quote (79 Bytes)	29
FB: Future Options Quote (80 Bytes)	30
FF: Futures Quote (71 Bytes)	31
FS: Strategy Quote (92 Bytes)	32
Market Depth Messages	33
H: Option Market Depth (Min 83 Bytes / Max 199 Bytes)	33
HB: Future Options Market Depth (Min 86 Bytes / Max 210 Bytes)	34
HF: Futures Market Depth (Min 77 Bytes / Max 201 Bytes)	36
HS: Strategy Market Depth (Min 105 Bytes / Max 230 Bytes)	38
Trade Cancellation Messages	39
I: Option Trade Cancellation (87 Bytes)	39
IB: Future Options Trade Cancellation (78 Bytes)	40
IF: Futures Trade Cancellation (79 Bytes)	41
IS: Strategy Trade Cancellation (88 Bytes)	41
Instrument Keys Messages	42
J: Option Instrument Keys (178 Bytes)	42
JB: Future Options Instrument Keys (179 Bytes)	44
JE: Underlying Instrument Keys (69 Bytes)	46
JF: Futures Instrument Keys (181 Bytes)	46
JS: Strategy Instrument Keys (Min 202 Bytes / Max 962 Bytes)	48
Summary Messages	50
N: Option Summary (161 Bytes)	50
NB: Future Options Summary (160 Bytes)	53
NF: Futures Summary (157 Bytes)	55
NS: Strategy Summary (145 Bytes)	58
Beginning of Summary Messages	60
Q: Beginning of Options Summary (33 Bytes)	60
QB: Beginning of Future Options Summary (33 Bytes)	60
QF: Beginning of Futures Summary (33 Bytes)	60
QS: Beginning of Strategy Summary (33 Bytes)	60
Trade Correction Messages	61
XF: Futures Trade Correction (78 Bytes)	61
Other Messages	62
GR: Group Status (40 Bytes)	62
GS: Group Status (Strategies) (36 Bytes)	62
KF: Future Deliverables (min 78 / max 686 Bytes)	62
L: Bulletins (114 Bytes)	63
S: End of Sales (39 Bytes)	64
TT: Tick Table (Min 103 Bytes / Max 567 Bytes)	64
U: End of Transmission (39 Bytes)	65

V: Circuit Assurance (38 Bytes)	65
SD: Start of day (41 Bytes)	66
Fields Description	67
Price and Fraction Indicator Code	67
Options Markers	67
Status Markers	68
Price Indicator Markers	69
Indicator Code	70
Currency Codes	70
Line Definitions	70
Error Definitions	71
Protocol Version	71
Market Flow Indicators	71
Month Codes	72
Tick Table	73
Strategy Code	73
Strategy Type	75
Processing Strategies	76
Messages to Use	76
Process	76

Section 1 Introduction

The Montréal Exchange Inc. (MX) High Speed Vendor Feed (HSVF) User Datagram Protocol (UDP) Multicast was developed by the Information Technology (IT) division of the Montréal Exchange Inc. (MX), a member of the TMX Group Inc.

The HSVF UDP Multicast is comprised of trades, quotes, market depth, strategies, bulletins, summaries and other statistics. Information is provided on all MX listings.

The UDP provides to the HSVF Participant a faster dissemination flow of messages. HSVF Participants use UDP lines to obtain the Market Dissemination flow from HSVF Repeaters. Each UDP line contains a specific market, a specific market depth and a specific protocol version.

1.1 Objective

The main objective of the **MX SOLA® HSVF UDP Multicast Specifications Guide** is to provide information to HSVF Participant in the functional design of their application intended to receive the HSVF feed.

1.2 Scope

This **MX SOLA® HSVF UDP Multicast Specifications Guide** defines the communications interface and message formats for the high speed transmission which broadcasts real-time trading and statistical information from Bourse de Montréal Inc.

1.3 Products Information

Detailed and up-to-date information regarding MX products is available on the MX Website at http://www.m-x.ca/accueil_en.php under **MX Products**.

For detailed information regarding expiry cycles, refer to:
http://www.m-x.ca/f_publications_en/cycles_opt_en.pdf.

1.4 Technical Support

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E-mail: marketops@tmx.com

Section 2 Trading Overview

2.1 Connectivity

All messages which comprise the MX-HSVF are transmitted to the user on a dedicated line. Each message type is fixed in format. Re-transmission of any data is available on the transmission line.

2.2 Interface

MX broadcasts the HSVF feed using both the UDP and TCP/IP broadcast interface as follows:

- Real-time market dissemination flow is broadcasted according to a defined timeline using the UDP interface to allow Participants to connect.
- The TCP interface retransmission can be used by the Participants to perform queries of missing messages of the UDP feed.

2.3 HSVF Feed Message Sequence of a Typical Day

During a typical day, all messages that comprise the MX-HSVF are transmitted following the schedule illustrated below. Participants can connect at **Start of Connection** time, which is when the broadcast starts and the tick table and the dictionary are sent out. Up to the opening of the market, the information regarding the instrument keys, summaries, quote/market depth is broadcasted. At the opening of the market, trades and trade cancellations resulting from the market activities are transmitted. The connection ends at **End of Connection** time after the market closure.

Table 1: Information Broadcast at Each Trading Phase

Trading Phases/ Information Broadcasted	Other Messages	Instrument Dictionary	Summary	Quote / Market Depth	Trade / Trade Cancellation
Start of Connection	X				
Tick Table / Dictionary / Initialization	X	X	X	X	X
Pre-opening	X	X	X	X	
Opening	X	X	X	X	X

Surveillance Intervention	X	X	X	X	X
End of Day	X	X	X	X	X
End of Connection	X				

GTE Test Environment

MX provides a GTE test environment to test the connection to HSVF with the pre-determined schedule. Please contact MX's Technical Help Desk for more details.

2.4 Transmission Format

A UDP packet can contain multiple HSVF messages. The UDP packet is built as follows:

UDP Packet			
HSVF Message 1	HSVF Message 2		HSVF Message N

A packet can have a maximum of 1000 characters.

Each message is framed by an STX and an ETX character. The format used is:

HSVF Message			
STX	Message Header	Message Body	ETX

STX and ETX indicate the beginning and the end of the record being transmitted.

2.5 Data Format

Each message consists of a message header followed by the message body, which varies in format according to the message type.

Two different message header exists:

- Message Header: used in LI, LO, KI, KO, ER, RT, RB, RE
- Long Message Header: used in all other messages

Table 2: Message Header

Field Name	L	T	Definition / Validation Rules
Sequence Number	10	N	Each message is assigned a sequence number starting at '0000000001' every day and incremented by 1 for each message sent. Note: Message sequencing is per line. There is no validation of message sequence for incoming messages. The sequence numbers will range from 0000000001 to 9999999999 (decimal, ASCII). Retransmitted messages will contain the original sequence numbers.
Message Type	2	X	Identifies the type of message being sent. Format is left-aligned, right 'blank' filled (if necessary).

Table 3: Long Message Header

Field Name	L	T	Definition / Validation Rules
Sequence Number	10	N	Each message is assigned a sequence number starting at '0000000001' every day and incremented by 1 for each message sent. Note: Message sequencing is per line. There is no validation of message sequence for incoming messages. The sequence numbers will range from 0000000001 to 9999999999 (decimal, ASCII). Retransmitted messages will contain the original sequence numbers.
Message Type	2	X	Identifies the type of message being sent. Format is left-aligned, right 'blank' filled (if necessary).
Timestamp	20	N	System timestamp, YYYYMMDDHHMMSSmmmmuuu.

2.6 TCP Retransmission Capability

Action	Particulars
Normal Connection at Start of Day	- Participant connects to the specified port - Participant sends a LI: Login message ex. 0000000001LI[User][Pwd][Timestamp][Protocol] - If the login is rejected, MX sends ER: Error Message to the Participant - If the login is accepted, MX sends KI: Login Acknowledgement message to the Participant Note: For more details on protocol versions, refer to Section 4.9 - Protocol Version
Retransmission (From a specific sequence number of a specific interval of messages)	- Participant sends a RT: Retransmission Request message ex. 0000013262RTO200000132470000013259 - If the transmission request is accepted, MX sends a RB: Retransmission Begin message to the Participant - MX sends all messages with sequence numbers greater than '0000013247' up to and including '0000013259' for the line - MX sends a RE: Retransmission End message to the Participant Note: If the Exchange sequence number is lower than the Start sequence number, the transmission request is rejected by a ER: Error Message Note: If the Start sequence number is higher than the End sequence number, the transmission request is rejected by a ER: Error Message
Disconnection	- Participant sends LO:Logout message ex. 0000000001LO - If the logout is rejected, MX sends ER: Error Message to the Participant - If the logout is accepted, MX sends KO: Logout Acknowledgement message to the Participant

2.7 Intraday Failover to Disaster Recovery Site

In case of major failure of the trading environment located at the primary site, trading will resume from the Disaster Recovery site.

From the DR site, HSVF will start as a new day:

- Message Sequence Number will be reset to 1.
- First message disseminated will be a Bulletin indicating sequence numbers are resetting.
- Trading statistics will be reset to 0 (volume, open, high, low).
- Pre-DR trading statistics will NOT be broadcasted.
- A complete instrument dictionary will be disseminated.
- A new start of day summary will be disseminated.
- Information from the pre-DR session will be unavailable.

Section 3 Messages

3.1 Message Types

This section includes a list of all message types grouped within categories.

Note: HSVF users must have the ability to skip and ignore any message that is not defined below. MX may introduce new message types to support extended functions in the future. Because new message types may be defined in future versions of the protocol, anyone using this version of the HSVF protocol must be able to avoid impact of undefined new messages types they may receive.

Section 3.3 - TCP (Retransmission) Messages	
3.3.1	LI: Login
3.3.2	LO: Logout
3.3.3	KI: Login Acknowledgement
3.3.4	KO: Logout Acknowledgement
3.3.5	RT: Retransmission Request
3.3.6	RB: Retransmission Begin
3.3.7	RE: Retransmission End
3.3.8	ER: Error Message
Section 3.4 - Trade Messages	
3.4.1	C: Option Trade
3.4.2	CB: Future Options Trade
3.4.3	CF: Future Trade
3.4.4	CS: Strategy Trade
Section 3.5 - Request for Quote (RFQ) Messages	
3.5.1	D: Option Request for Quote
3.5.2	DB: Future Option Request for Quote
3.5.3	DF: Future Request for Quote

3.5.4	DS: Strategy Request for Quote
Section 3.6 - Instrument Schedule Notice	
3.6.1	E: Instrument Schedule Notice Option
3.6.2	EB: Instrument Schedule Notice Futures Option
3.6.3	EF: Instrument Schedule Notice Future
3.6.4	ES: Instrument Schedule Notice Strategy
Section 3.7 - Quote Messages	
3.7.1	F: Option Quote
3.7.2	FB: Future Options Quote
3.7.3	FF: Future Quote
3.7.4	FS: Strategy Quote
Section 3.8 - Market Depth Messages	
3.8.1	H: Option Market Depth
3.8.2	HB: Future Options Market Depth
3.8.3	HF: Future Market Depth
3.8.4	HS: Strategy Market Depth
Section 3.9 - Trade Cancellation Messages	
3.9.1	I: Option Trade Cancellation
3.9.2	IB: Future Options Trade Cancellation
3.9.3	IF: Future Trade Cancellation
3.9.4	IS: Strategy Trade Cancellation
Section 3.10 - Instrument Keys Messages	
3.10.1	J: Options Instrument Keys
3.10.2	JB: Future Options Instrument Keys
3.10.3	JE: Underlying Instrument Keys

3.10.4	JF: Future Instrument Keys
3.10.5	JS: Strategy Instrument Keys
Section 3.11 - Summary Messages	
3.11.1	N: Option Summary
3.11.2	NB: Future Options Summary
3.11.3	NF: Future Summary
3.11.4	NS: Strategy Summary
Summary messages will be sent: • At the beginning of the day to define the instruments traded on that day • After a trade cancellation (Open/High/Low/Last) • At the end of the day with relevant data such as the Open/High/Low/Last/Volume • During the day if a new instrument is added	
Section 3.12 - Beginning of Summary Messages	
3.12.1	Q: Beginning of Option Summary
3.12.2	QB: Beginning of Future Options Summary
3.12.3	QF: Beginning of Futures Summary
3.12.4	QS: Beginning of Strategy Summary
Section 3.13 – Trade Correction Messages	
3.13.1	XF: Futures Trade Correction
Section 3.14 - Other Messages	
3.14.1	GR: Group Status
3.14.2	GS: Group Status (Strategies)
3.14.3	KF: Future Deliverables
3.14.4	L: Bulletins
3.14.5	S: End of Sales
3.14.6	TT: Tick Table
3.14.7	U: End of Transmission
3.14.8	V: Circuit Assurance ("Heartbeat")
3.14.9	SD: Start of day

3.2 Conventions

In the following tables, the **L** column represents the length in bytes of the described field, and the **T** column ('Data Type') will be represented by the following characters (**A** = Alphabetic, **N** = Numeric, **X** = Alphanumeric).

- Whenever a field is indicated as being blank, it contains the ASCII space character (hex 20).
- **Alphabetic Fields – A:**
Letters (A to Z) left justified, blank filled unless stated otherwise.
- **Numeric Fields – N:**
Numbers (0 to 9), right justified, zero filled with a possibility to see a '.' (ASCII character hex 2).
- **Alphanumeric Fields – X:**
All characters possible (numbers, letters, others), right justified, zero filled, with the exception of the following fields, which are left justified, and blank filled:
 - Instrument External Code
 - Root Symbol (Options related messages)
 - Symbol (Strategy related messages)

3.3 TCP (Retransmission) Messages

3.3.1 LI: Login (52 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header
User ¹	16	X	Participant logging in
Pwd ¹	16	X	Participant password
Timestamp	6	N	Time submitted
Protocol	2	X	Refer to Protocol Version

3.3.2 LO: Logout (12 Bytes)

Field Name	L	T	Definition / Validation Rules
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¹ For future use

Message Header	12	X	Refer to Message Header
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3.3.3 KI: Login Acknowledgement (12 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header

3.3.4 KO: Logout Acknowledgement (12 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header

3.3.5 RT: Retransmission Request (34 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header
Line	2	X	Specific address and port on which market is disseminated based on a list of CPUs and Market Depth Setting Refer to Line Definitions
Start	10	N	Starting message number
End	10	N	Ending message number

3.3.6 RB: Retransmission Begin (12 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header

3.3.7 RE: Retransmission End (12 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header

3.3.8 ER: Error Message (96 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	12	X	Refer to Message Header
ErrorCode	4	N	Error Code
ErrorMsg	80	X	Error Message

3.4 Trade Messages

3.4.1 C: Option Trade (86 Bytes)

Field Name		L	T	Definition / Validation Rules
Long Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol (symbol of the underlying)
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Volume		8	N	Number of contracts for the trade Refer to Indicator Code
Trade Price		7	N	Price at which the transaction took place
Trade Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Net Change Sign +/-		1	A	For the net change field
Net Change		7	N	Net change = last trade price - previous close
Net Change Fraction Indicator		1	X	Fraction indicator for the net change price Refer to Fraction Indicator Code
Price Indicator Marker		1	A	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number		8	X	Unique Trade Number for this instrument

3.4.2 CB: Future Options Trade (87 Bytes)

Field Name	L	T	Definition / Validation Rules
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Long Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call / Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Volume		8	N	Total number of contracts traded Refer to Indicator Code
Trade Price		7	N	Price at which the transaction took place
Trade Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Price Indicator Marker		1	A	Identifies the type of transaction Refer to Price Indicator Markers
Net Change Sign +/-		1	A	For net change field (sign)
Net Change		7	N	Net change = last trade price - previous settlement price
Net Change Fraction Indicator		1	A	Fraction indicator for the net change Refer to Fraction Indicator Code
Trade Number		8	X	Unique Trade Number for this instrument

Before the closing settlement prices are determined and transmitted for all OGB (Options on the 10-Year Canadian Government Bond Futures) and OBX (Options on the 3-Month Canadian Bankers' Acceptance Futures), most fields for each series will be blank/zero filled except for the Price field, which will contain the closing settlement price.

3.4.3 CF: Futures Trade (78 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Volume		8	N	Total number of contracts traded Refer to Indicator Code
Trade Price		7	N	Price at which the transaction took place
Trade Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Net Change Sign +/-		1	X	For net change field (sign)
Net Change		7	N	Net change = last trade price - previous settlement price
Net Change Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Price Indicator Marker		1	X	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number		8	X	Unique Trade Number for this instrument

Before the closing settlement prices are determined and transmitted for all CGB (10-Year Canadian Government Bond Futures) and BAX (3-Month Canadian Bankers' Acceptance Futures) products, most fields for each series will be blank/zero filled except for the Price field, which will contain the closing settlement price.

3.4.4 CS: Strategy Trade (98 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:

Exchange ID	1	A	Exchange on which the trade occurred Q = Montreal
Symbol	30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with ".", "+", "-"
Volume	8	N	Total number of contracts traded Refer to Indicator Code
Trade Price Sign +/-	1	X	For Trade Price field (sign)
Trade Price	7	N	Price at which the transaction took place
Trade Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Net Change Sign +/-	1	X	For net change field
Net Change	7	N	Net change = last trade price - previous close
Net Change Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Price Indicator Marker	1	X	Identifies type of transaction Refer to Price Indicator Markers
Trade Number	8	X	Unique Trade Number for this instrument

3.5 Request for Quote (RFQ) Messages

3.5.1 D: Option Request for Quote (RFQ) (61 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Requested Size		8	X	Size of the market requested Refer to Indicator Code
Requested Market Side		1	X	Requested quote side: B = Buy S = Sell 2 = Both

3.5.2 DB: Future Options Request for Quote (RFQ) (62 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions. Refer to Fraction Indicator Code
Requested Size		8	X	Size of the market requested Refer to Indicator Code
Requested Market Side		1	X	Requested quote side: B = Buy S = Sell 2 = Both

3.5.3 DF: Futures Request for Quote (RFQ) (53 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Base symbol for the contract
	Symbol Month	1	A	Symbol month for the contract Refer to Month Codes
	Symbol Year	2	N	Last two digits of the contract symbol year
	Expiry Day	2	N	Expiry day of the contract
Requested Size		8	X	Size of the market requested Refer to Indicator Code
Requested Market Side		1	X	Requested quote side: B = Buy S = Sell 2 = Both

3.5.4 DS: Strategy Request for Quote (RFQ) (72 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
Symbol		30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with ".", "+", "-"
Requested Size		8	X	Size of the market requested Refer to Indicator Code
Requested Market Side		1	X	Requested quote side: B = Buy S = Sell 2 = Both

3.6 Instrument Schedule Notice Messages

The E, EB, EF, ES messages indicate the opening hour for their respective instrument.

3.6.1 E: Instrument Schedule Notice Option (59 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol (symbol of the underlying)
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	X	Option strike Price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Series Status		1	A	Series status of the trading instrument Refer to Status Markers
Scheduled Status Change Time		6	N	Time at which the status change is scheduled, HHMMSS

3.6.2 EB: Instrument Schedule Notice Futures Option (60 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year

	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	S	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines the number of decimal places or fraction positions Refer to Fraction Indicator Code
Series Status		1	A	Series status of the trading instrument Refer to Status Markers
Scheduled Status Change Time		6	N	Time at which the status change is scheduled, HHMMSS

3.6.3 EF: Instrument Schedule Notice Future (51 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Series Status		1	A	Series status of the trading instrument Refer to Status Markers
Scheduled Status Change Time		6	N	Time at which the status change is scheduled, HHMMSS

3.6.4 ES: Instrument Schedule Notice Strategy (70 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
Strategy Symbol		30	X	Identifies the strategy The legs are defined in message type JS

			Alphanumeric with ".", "+", "-"
Series Status	1	A	Series status of the trading instrument Refer to Status Markers
Scheduled Status Change Time	6	N	Time at which the status change is scheduled, HHMMSS

3.7 Quote Messages

3.7.1 F: Option Quote (79 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the quote occurred Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Bid Price		7	X	Bid price for the option series
Bid Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Number of option contracts represented by the Bid Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Ask Price		7	X	Ask price for the option series
Ask Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

Ask Size	5	X	Number of option contracts represented by the Ask Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Instrument Status Marker	1	A	Indicates instrument status Refer to Status Markers

3.7.2 FB: Future Options Quote (80 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the quote occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Price		7	X	Bid price for the series
Bid Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Total number of contracts being bid at this price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Ask Price		7	X	Ask price for the series

Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Ask Size	5	X	Total number of contracts being offered at this price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Instrument Status Marker	1	A	Indicates instrument status Refer to Status Markers

3.7.3 FF: Futures Quote (71 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Bid Price		7	X	Bid price for the future contract
Bid Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Number of futures contracts represented by the Bid Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Ask Price		7	X	Ask Price for the future contract
Ask Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

Ask Size	5	X	The number of futures contracts represented by the Ask Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Instrument Status Marker	1	A	Indicates instrument status Refer to Status Markers

3.7.4 FS: Strategy Quote (92 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal
Symbol	30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with ".", "+", "-"
Bid Price Sign +/-	1	X	For Bid Price field
Bid Price	7	X	Bid Price for the future contract
Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size	5	X	Number of futures contracts represented by the Bid Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Ask Price sign +/-	1	X	For Ask Price field
Ask Price	7	X	Ask price for the future contract
Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Ask Size	5	X	The number of futures contracts represented by the Ask Price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
Instrument Status Marker	1	X	Indicates instrument status

			Refer to Status Markers
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3.8 Market Depth Messages

3.8.1 H: Option Market Depth (Min 83 Bytes / Max 199 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Instrument Status Marker		1	A	Instrument status Refer to Status Markers
Number of Level		1	N	Number of level for the trading instrument 1 to 5
U p t o 5 t i m e s	Level of Market Depth	1	X	Level of market depth 1 to 5 and A (Implied)
	Bid Price	7	X	Bid price for the option series For Implied, it represents the best (1st limit) indicative implied bid price
	Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

	Bid Size	5	X	Number of option contracts represented by the Bid Price For Implied, it represents the indicative quantity at the best (1st limit) implied bid price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Bid Orders	2	X	Number of bid orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied bid orders making up the implied bid size at that implied bid price If greater than 99-> the 2nd character becomes an exponent. Refer to Indicator Code
	Ask Price	7	X	Ask price for the option series For Implied, it represents the best (1st limit) indicative implied ask price
	Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Ask Size	5	X	Number of option contracts represented by the Ask Price For Implied, it represents the indicative quantity at the best (1st limit) implied ask price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Ask Orders	2	X	Number of Ask Orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied ask orders making up the implied ask size at that implied ask price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code

3.8.2 HB: Future Options Market Depth (Min 86 Bytes / Max 210 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3 :
Exchange ID	1	A	Identifies the exchange Q = Montreal

S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Instrument Status Marker		1	A	Instrument status Refer to Status Markers
Number of Level		1	N	Number of level for the trading instrument, 1 to 5
U p t o 5 t i m e s	Level of Market Depth	1	X	Level of market depth 1 to 5 and A (Implied)
	Bid Price	7	X	Bid Price for the series. For Implied, it represents the best (1st limit) indicative implied bid price
	Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Bid Size	5	X	Total number of contracts being bid at this price For Implied, it represents the indicative quantity at the best (1st limit) implied bid price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Bid Orders	2	X	Number of bid orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied bid orders making up the implied bid size at that implied bid price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code

	Ask Price	7	X	Ask price for the series For Implied, it represents the best (1st limit) indicative implied ask price
	Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Ask Size	5	X	Total number of contracts being offered at this price For Implied, it represents the indicative quantity at the best (1st limit) implied ask price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Ask Orders	2	X	Number of Ask Orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied ask orders making up the implied ask size at that implied ask price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code

3.8.3 HF: Futures Market Depth (Min 77 Bytes / Max 201 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
Symbol	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Instrument Status Marker		1	A	Instrument status Refer to Status Markers
Number of Level		1	N	Number of level for the trading instrument, 1 – 5
Up to 5 ti	Level of Market Depth	1	X	Level of market depth, 1 – 5 and A (Implied)

m e s	Bid Price	7	X	Bid price for the future contract For Implied, it represents the best (1st limit) indicative implied bid price
	Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions. Refer to Fraction Indicator Code
	Bid Size	5	X	Number of futures contracts represented by the Bid Price For Implied, it represents the indicative quantity at the best (1st limit) implied bid price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Bid Orders	2	X	Number of Bid Orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied bid orders making up the implied bid size at that implied bid price If greater than 99-> the 2nd character becomes an exponent
	Ask Price	7	N	Ask Price for the future contract For Implied, it represents the best (1st limit) indicative implied ask price
	Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Ask Size	5	X	The number of futures contracts represented by the Ask Price For Implied, it represents the indicative quantity at the best (1st limit) implied ask price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Ask Orders	2	X	Number of Ask Orders, present at a given moment in the order book For Implied, it represents the indicative number of implied ask orders making up the implied ask size at that implied ask price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code

3.8.4 HS: Strategy Market Depth (Min 105 Bytes / Max 230 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
Symbol		30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with ".", "+", "-"
Instrument Status Marker		1	A	Instrument status Refer to Status Markers
Number of Level		1	N	Number of level for the trading instrument, 1 – 5
U p to 5 t i m e s	Level of Market Depth	1	X	Level of market depth, 1 – 5 and A (Implied) For the Bid Price field
	Bid Price Sign +/-	1	X	
	Bid Price	7	X	Bid price for the strategy instrument For Implied, it represents the best (1st limit) indicative implied bid price
	Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Bid Size	5	X	Number of strategy units represented by the Bid Price For Implied, it represents the indicative quantity at the best (1st limit) implied bid price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Bid Orders	2	X	Number of Bid Orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied bid orders making up the implied bid size at that implied bid price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code
Ask Price Sign +/-		1	X	For the Ask Price field

	Ask Price	7	X	Ask price for the strategy instrument For Implied, it represents the best (1st limit) indicative implied ask price
	Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Ask Size	5	X	The number of strategy units represented by the Ask Price For Implied, it represents the indicative quantity at the best (1st limit) implied ask price If size is greater than 99999, the 5th character becomes an exponent Refer to Indicator Code
	Number of Ask Orders	2	X	Number of Ask Orders, present at a given moment, in the order book For Implied, it represents the indicative number of implied ask orders making up the implied ask size at that implied ask price If greater than 99-> the 2nd character becomes an exponent Refer to Indicator Code

3.9 Trade Cancellation Messages

3.9.1 I: Option Trade Cancellation (87 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day

Volume	8	N	Number of contracts being cancelled Refer to Indicator Code
Trade Price	7	N	Price at which the transaction took place
Trade Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Price Indicator Marker	1	A	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number	8	X	Unique trade Number for this instrument

A cancellation will reduce the total volume, value and transactions by the amount of the cancelled trade. A cancellation message is followed by a **N: Option Summary** which will reflect the corrected market.

3.9.2 IB: Future Options Trade Cancellation (78 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Volume		8	N	Total number of contracts traded Refer to Indicator Code
Price		7	N	Price at which the transaction took place
Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

Price Indicator Marker	1	A	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number	8	X	Unique trade Number for this instrument

A cancellation will reduce the total volume, value and transactions by the amount of the cancelled trade. A cancellation message is followed by a **NB: Future Options Summary** message which will reflect the corrected market.

3.9.3 IF: Futures Trade Cancellation (79 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
Symbol	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Volume		8	N	Number of contracts being cancelled Refer to Indicator Code
Trade Price		7	N	Estimated price at which the transaction took place
Trade Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Price Indicator Marker		1	X	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number		8	X	Unique trade Number for this instrument

A cancellation will reduce the total volume, value and transactions by the amount of the cancelled trade. A cancellation message is followed by a **NF: Future Summary message** which will reflect the corrected market.

3.9.4 IS: Strategy Trade Cancellation (88 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:

Exchange ID	1	A	Exchange on which the trade occurred Q = Montreal
Symbol	30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with ".", "+", "-"
Volume	8	N	Number of contracts being cancelled Refer to Indicator Code
Trade Price sign +/-	1	X	For the Trade Price field
Trade Price	7	N	Estimated price at which the transaction took place
Trade Price Fraction Indicator	1	X	Defines the number of decimal places or fraction positions Refer to Fraction Indicator Code
Trade Number	8	X	Unique trade Number for this instrument

A cancellation will reduce the total volume, value and transactions by the amount of the cancelled trade. A cancellation message is followed by a **NS: Strategy Summary** message which will reflect the corrected market.

3.10 Instrument Keys Messages

3.10.1 J: Option Instrument Keys (178 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes
	Strike Price	7	N	Option strike price in full length Refer to Tick Table
	Strike Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day

Strike Price Currency	3	A	Currency used for the option Strike Price Refer to Currency Codes
Maximum Number of Contracts per Order	6	X	Maximum authorized number of contract per order Refer to Indicator Code
Minimum Number of Contracts per Order	6	X	Minimum authorized number of contract per order Refer to Indicator Code
Maximum Threshold Price	7	X	Maximum threshold price authorized for an option contract Refer to Indicator Code
Maximum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Minimum Threshold Price	7	X	Minimum threshold price authorized for an option contract Refer to Indicator Code
Minimum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Tick Increment	7	X	Contains the Tick Table Short Name if available. If so, field starts with TT= Otherwise, indicates precision with which the price of an order limit can be expressed Refer to Tick Table
Tick Increment Fraction Indicator	1	X	If Tick Increment starts with TT= (Table Name available), the Fraction Indicator is blank. Otherwise, it defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Option type	1	A	Type of option: A = American E = European
Market Flow Indicator	2	X	Defines the type of instruments Refer to Market Flow Indicators
Group Instrument	2	X	Group of the instrument
Instrument	4	X	Instrument
Instrument External Code	30	X	External identifier used by traders when entering an order
Option Marker	2	A	Refer to Options Markers

Underlying Symbol Root	12	X	Symbol root for the underlying security
Contract Size	8	N	Defines the quantity of an underlying per contract
Tick Value	7	N	Exchange premium multiplier for derivatives with a contract size equal to 1. All derivatives with a contract size > 1 will have a tick value defaulted to 1.
Tick Value Fraction Indicator	1	X	Defines number of decimal places or fraction positions
Currency	3	A	Defines the currency of an underlying Refer to Currency Codes
Delivery Type	1	X	Indicates the delivery type of the instrument C = Cash, P = Physical
Last Trading Date Time	14	N	Indicates when the instrument stops trading, YYYYMMDDHHMMSS

3.10.2 JB: Future Options Instrument Keys (179 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call / Put Code	1	A	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Fraction indicator for the strike price Refer to Fraction Indicator Code
Expiry Date		6	N	Expiry Date, YYYYMMDD
Strike Price Currency		3	A	Currency used for the option strike price Refer to Currency Codes

Maximum Number of Contracts per Order	6	X	Maximum authorized number of contract per order Refer to Indicator Code
Minimum Number of Contracts per Order	6	X	Minimum authorized number of contract per order Refer to Indicator Code
Maximum Threshold Price	7	X	Maximum threshold price authorized for an option contract Refer to Indicator Code
Maximum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Minimum Threshold Price	7	X	Minimum threshold price authorized for an option contract Refer to Indicator Code
Minimum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Tick Increment	7	X	Contains the Tick Table Short Name if available. If so, field starts with TT= Otherwise, it is the precision with which the price of an order limit can be expressed Refer to Tick Table
Tick Increment Fraction Indicator	1	X	If Tick Increment starts with TT= (Table Name available), Fraction Indicator is blank. Otherwise, it defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Market Flow Indicator	2	X	Defines the type of instruments Refer to Market Flow Indicators
Group Instrument	2	X	Group of the instrument
Instrument	4	X	Instrument
Instrument External Code	30	X	External identifier used by traders when entering an order
Contract Size	8	N	Defines the quantity of an underlying per contract
Tick Value	7	N	Exchange premium multiplier for derivatives with a contract size equal to 1 All derivatives with a contract size > 1 will have a tick value defaulted to 1

Tick Value Fraction Indicator	1	X	Defines number of decimal places or fraction positions
Currency	3	A	Defines the currency of an underlying Refer to Currency Codes
Delivery Type	1	A	Indicates the delivery type of the instrument C = Cash P = Physical
Underlying Root Symbol	6	A	Root symbol of the underlying future
Underlying Symbol Month	1	A	Symbol month of the underlying futures contract
Underlying Symbol Year	2	N	Last 2 digits of the symbol year of the underlying futures contract
Last Trading Date Time	14	N	Indicates when the instrument stops trading, YYYYMMDDHHMMSS

3.10.3 JE: Underlying Instrument Keys (69 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Exchange on which the trade occurred. Q = Montreal
Group Instrument	2	X	Group of the instrument
Instrument	4	X	Instrument
Instrument External Code	30	X	External identifier used by traders when entering an order

The 'Underlying Instrument Keys' message will be the first messages sent during the day.

There is no 'Underlying Summary' message attached to the 'Instrument Keys' messages

3.10.4 JF: Futures Instrument Keys (181 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Exchange on which the trade occurred Q = Montreal

S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Expiry Date		6	N	Expiry Date, YYMMDD
Maximum Number of Contracts per Order		6	X	Maximum authorized number of contract per order Refer to Indicator Code
Minimum Number of Contracts per Order		6	X	Minimum authorized number of contract per order Refer to Indicator Code
Maximum Threshold Price		7	N	Maximum threshold price authorized for an option contract Refer to Indicator Code
Maximum Threshold Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Minimum Threshold Price		7	N	Minimum threshold price authorized for an option contract Refer to Indicator Code
Minimum Threshold Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Tick Increment		7	X	Contains the Tick Table Short Name if available. If so, field starts with TT= Otherwise, it is precision with which the price of an order limit can be expressed Refer to Tick Table
Tick Increment Fraction Indicator		1	X	If Tick Increment starts with TT= (Table Short Name available), Fraction Indicator is blank Otherwise, it defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Market Flow Indicator		2	X	Defines the type of instruments Refer to Market Flow Indicators
Group Instrument		2	X	Group of the instrument
Instrument		4	X	Instrument

Instrument External Code		30	X	External identifier used by traders when entering an order
Contract Size		8	N	Defines the quantity of an underlying per contract
Tick Value		7	N	Exchange premium multiplier for derivatives with a contract size equal to 1. All derivatives with a contract size > 1 will have a tick value defaulted to 1.
Tick Value Fraction Indicator		1	X	Defines number of decimal places or fraction positions
Currency		3	A	Defines the currency of an underlying Refer to Currency Codes
Underlying Symbol		12	X	Symbol root for the underlying security
Delivery Type		1	A	Indicates the delivery type of the instrument. C = Cash P = Physical
Associated Product	Root Symbol	6	A	Associated product's class symbol
	Symbol Month	1	A	Symbol month
	Symbol Year	2	N	Last two digits of the symbol year
	Expiry Day	2	N	Expiry day
Last Trading Date Time		14	N	Indicates when the instrument stops trading, YYYYMMDDHHMMSS

3.10.5 JS: Strategy Instrument Keys (Min 202 Bytes / Max 962 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Exchange on which the trade occurred Q = Montreal
Symbol	Strategy Symbol	30	X	Identification of the strategy Alphanumeric with ".", "+", "-", "
Expiry Year		2	N	Last two digits of the contract expiry year from the first expiring leg
Expiry Month		1	A	Contract expiry month from the first expiring leg

			Refer to Month Codes
Expiry Day	2	N	Contract expiry day from the first expiring leg
Maximum Number of Contracts per Order	6	X	Maximum authorized number of contract per order Refer to Indicator Code
Minimum Number of Contracts per Order	6	X	Minimum authorized number of contract per order Refer to Indicator Code
Maximum Threshold Price	7	X	Maximum threshold price authorized for an option contract Refer to Indicator Code
Maximum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Minimum Threshold Price	7	X	Minimum threshold price authorized for an option contract Refer to Indicator Code
Minimum Threshold Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Tick Increment	7	X	Contains the Tick Table Short Name if available. If so, field starts with TT= Otherwise, it is the precision with which the price of an order limit can be expressed Refer to Tick Table
Tick increment Fraction Indicator	1	X	If Tick Increment starts with TT= (Table Short Name available), Fraction Indicator is blank Otherwise, it defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Market Flow Indicator	2	X	Defines the type of instruments Refer to Market Flow Indicators
Group Instrument	2	X	Group of the instrument
Instrument	4	X	Instrument
Instrument External Code	30	X	External identifier used by traders when entering an order
Strategy Allow Implied	1	A	Indicates if the Strategy supports Implied Pricing Y = Yes N = No

Strategy Code		2	X	Defines the code of the strategy. Refer to Section 0
Strategy Type		1	X	Defines the type of the strategy, “C” Classic “S” Strip “V” Covered Regular “R” Covered reference Fixed
Last Trading Date Time		14	N	Indicates when the instrument stops trading, YYYYMMDDHHMMSS
Number of Legs		2	N	Number of legs in the strategy, 2 to 40
From 2 to 40 times	Leg Group Instrument	2	X	Group of the Leg Instrument
	Leg Instrument	4	X	Leg Instrument
	Leg Ratio or Delta	4	N	Quantity (bought or sold) on underlying in the strategy Ratio: Value range of 1 to 9999 Delta: Value range of 0.01% to 9999%
	Leg Ratio or Delta Fraction Indicator	1	X	Defines number of decimal places or fraction positions (including + or - sign, + indicate a Buy and - indicate a Sell) Refer to Fraction Indicator Code
	Leg Price	7	N	Significant if the leg is a fixed leg of a covered strategy. Refer to fraction indicator
	Leg Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions. Empty if a non fixed leg. Refer to Fraction Indicator Code

3.11 Summary Messages

3.11.1 N: Option Summary (161 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange Q = Montreal
Sym	Root Symbol	6	X	Option base symbol
	Expiry Month	1	A	Option expiry month code Refer to Month Codes

b ol	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines fraction indicator for the strike price Refer to Fraction Indicator Code
	Expiry Year	2	N	Last two digits of the option expiry year
	Expiry Day	2	N	Option expiry day
Bid Price		7	N	Closing or most recent bid price
Bid Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Number of contracts represented by the Bid Price. If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Ask Price		7	N	Closing or most recent ask price
Ask Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Ask Size		5	X	Number of contracts represented by the Ask Price If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Last Price		7	N	Most recent trade price
Last Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Open Interest		7	N	This field contains current outstanding number of contracts in the series Updated on a trade by trade basis Refer to Indicator Code
Open Interest Date		6	N	Open interest date, YYMMDD
Tick		1	X	Determined by the difference between last price and the previous different trade price '+' = uptick '-' = downtick
Volume		8	N	Total number of contracts traded or current volume if sent after a cancellation

Net Change Sign +/-	1	X	For net change field
Net Change	7	N	Net change = last trade price – previous settlement price Net change will be zero if the option did not trade on the last business day or did not trade today
Net Change Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Open Price	7	N	Price of the first trade of the day
Open Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
High Price	7	N	Highest trade price of the day or current high price if sent after a cancellation
High Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Low Price	7	N	Lowest trade price of the day or current low price if sent after a cancellation
Low Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Option Marker	2	A	Refer to Options Markers
Settlement Price	7	N	Settlement Price
Settlement Price Fraction Indicator	1	N	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Previous Settlement Price	7	N	Previous Settlement Price
Previous Settlement Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Reason	1	X	Indicates the reason of the summary message S = Start of Day E = End of Day U = Instrument New or Update C = Trade Cancellation

An option summary message is sent following an option trade cancellation. An option summary message is also sent each day at the start of the day in order to provide a

list of options which will be trading each day. At that point, all price fields with the exception of Last Price (closing from the previous day), will contain zero values.

Any option summary sent after the **Q: Beginning of Option Summary** message contains the list of trading instruments for the day (sent prior to market opening) or the summaries after the close of the market for Bourse de Montréal Inc. options.

3.11.2 NB: Future Options Summary (160 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Option base symbol
	Symbol Month	1	A	Option symbol month code Refer to Month Codes
	Symbol Year	2	N	Last two digits of the option symbol year
	Expiry Day	2	N	Option expiry day
	Call/Put Code	1	S	C = Call P = Put
	Strike Price	7	N	Option strike price in full length
	Strike Price Fraction Indicator	1	X	Defines the number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Price		7	N	Closing bid price or current bid if sent after a cancellation
Bid Price Fraction Indicator		1	X	Defines the number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Total number of contracts being bid at this price If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Ask Price		7	N	Closing Ask Price or current Ask Price if sent after a cancellation
Ask Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions

			Refer to Fraction Indicator Code
Ask Size	5	X	Total number of contracts being offered at this price If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Last Price	7	N	Most current price
Last Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Open Interest	7	N	Outstanding number of contracts in the series as of the previous day
Open Interest Date	6	N	Open interest date, YYMMDD
Tick	1	X	Calculated on the difference of the last sale price to the different previous last price '+' = uptick '-' = downtick
Volume	8	N	Total volume of contracts traded for this option series during the day or current volume if sent after a cancellation Refer to Indicator Code
Net Change Sign +/-	1	X	For net change field
Net Change	7	N	Net change = last trade price – previous settlement price If no previous settlement price (new series) then net change is zero
Net Change Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Opening Price	7	N	Opening price of the option series for the day
Opening Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
High Price	7	N	Highest trade price of the options series for the day or current high price if sent after a cancellation
High Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

Low Price	7	N	Lowest trade price of the option series for the day or current low price if sent after a cancellation
Low Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Settlement Price	7	N	Settlement Price
Settlement Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Previous Settlement Price	7	N	Previous Settlement Price Refer to Fraction Indicator Code
Previous Settlement Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Reason	1	X	Indicates the reason of the summary message S = Start of Day E = End of Day U = Instrument New or Update C = Trade Cancel

A Future Options Summary is sent following a future option trade cancellation. A Future options summary is also sent each day at the start of the day in order to provide a list of Future options which will be trading each day. At that point, all price fields with the exception of Previous Settlement Price (closing from previous day) will contain zero values.

Any Future Options Summary sent after the **QB: Beginning of Future Options Summary** message contains the list of trading instruments for the day (sent prior to market opening) or the summaries after the close of the market for Bourse de Montréal Inc. Future options.

3.11.3 NF: Futures Summary (157 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
Symbol	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes

	Symbol Year	2	N	Two last digit of the future symbol year
	Expiry Day	2	N	Future expiry day
Bid Price		7	N	Closing bid or most recent bid if sent after a cancellation
Bid Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size		5	X	Number of contracts represented by the Bid Price If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Ask Price		7	N	Closing Ask Price or most recent Ask Price if sent after a cancellation
Ask Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Ask Size		5	X	Number of contracts represented by the ask price If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Last Price		7	N	Last trade price for the contract or the current price if sent after a cancellation
Last Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Open Price		7	N	Price of the first trade of the day
Open Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
High Price		7	N	Highest trade price of the day or current high price if sent after a cancellation
High Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Low Price		7	N	Lowest trade price of the day or current low price if sent after a cancellation
Low Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions

			Refer to Fraction Indicator Code
Settlement Price	7	N	Closing settlement price for the contract 0 until market closes
Settlement Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Net Change Sign +/-	1	X	For net change field
Net Change	7	N	Net change = last Trade Price – previous Settlement Price If no previous settlement price (new series) then net change is zero
Net Change Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Volume	8	N	Total number of contracts traded Refer to Indicator Code
Previous Settlement	7	N	Settlement Price for the previous day
Previous Settlement Fraction Indicator	1	X	Fraction indicator for the previous Settlement Price
Open Interest	7	N	Previous day's outstanding number of contracts in the series Refer to Indicator Code
Open Interest Date	6	N	Open interest date, YYMMDD
Reason	1	X	Indicates the reason of the summary message S = Start of Day E = End of Day U = Instrument New or Update C = Trade Cancel
External Price at Source	7	N	Last price obtained from an external pricing source
External Price Fraction Indicator	1	X	Fraction indicator for last price obtained from external pricing source

A Futures Summary is sent following a Futures trade cancellation. A Futures summary is also sent each day at the start of the day in order to provide a list of Futures which will be trading each day. At that point, all price fields with the exception of previous settlement will contain zero values.

Any summary sent after the **QF: Beginning of Future Summary** message contains the list of trading instruments for the day (sent prior to market opening) or the summaries after the close of the market for Bourse de Montréal Inc. Futures.

3.11.4 NS: Strategy Summary (145 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3 :
Exchange ID	1	X	Identifies the exchange Q = Montreal
Strategy Symbol	30	X	Identifies the strategy The legs are defined in message type JS Alphanumeric with “.”,”+”,”-“
Bid Price sign +/-	1	X	For the Bid Price field
Bid Price	7	N	Closing bid or most recent bid if sent after a cancellation
Bid Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Bid Size	5	X	Number of contracts represented by the Bid Price. If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Ask Price Sign +/-	1	X	For the Ask Price field
Ask Price	7	N	Closing ask or most recent ask if sent after a cancellation
Ask Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Ask Size	5	X	Number of contracts represented by the Ask Price. If size is greater than 99999, the 5 th character becomes an exponent Refer to Indicator Code
Last Price Sign +/-	1	X	For the Last Price field
Last Price	7	N	Last Trade Price for the contract or the current price if sent after a cancellation
Last Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions

			Refer to Fraction Indicator Code
Open Price Sign +/-	1	X	For the Open Price field
Open Price	7	N	Price of the first trade of the day
Open Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
High Price Sign +/-	1	X	For the High Price field
High Price	7	N	Highest trade price of the day or current high price if sent after a cancellation
High Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Low Price Sign +/-	1	X	For the Low Price field
Low Price	7	N	Lowest Trade Price of the day or current low price if sent after a cancellation
Low Price Fraction Indicator	1	X	Defines number of decimal or fraction positions Refer to Fraction Indicator Code
Net Change Sign +/-	1	X	For net change field
Net Change	7	N	Net change = last trade price – previous close If no previous settlement price (new series) then net change is zero
Net Change Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
Volume	8	N	Total number of contracts traded Refer to Indicator Code
Reason	1	X	Indicates the reason of the summary message S = Start of Day E = End of Day U = Instrument New or Update C = Trade Cancel

A Strategy Summary is sent following a Strategy Trade Cancellation. A Strategy Summary is also sent each day at the start of the day in order to provide a list of Strategies which will be trading each day. At that point, all price fields will contain zero values.

Any Strategy Summary sent after the **QS: Beginning of Strategy Summary** message contains the list of trading instruments for the day (sent prior to market

opening) or the summaries after the close of the market for Bourse de Montréal Inc. Strategies.

3.12 Beginning of Summary Messages

3.12.1 Q: Beginning of Options Summary (33 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal

This message indicates that the beginning and the end of day **N: Option Summary** messages are to follow. Other messages (such as bulletins) can be interspersed with the summaries.

3.12.2 QB: Beginning of Future Options Summary (33 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal

This message indicates that the beginning and the end of day **NB: Future Options Summary** messages are to follow. Other messages (such as bulletins) can be interspersed with the summaries.

3.12.3 QF: Beginning of Futures Summary (33 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal

This message indicates that the beginning or end-of-day **NF: Future Summary** messages are to follow. Other messages can be interspersed with the summaries.

3.12.4 QS: Beginning of Strategy Summary (33 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:

Exchange ID	1	A	Identifies the exchange Q = Montreal
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This message indicates that the beginning or the end of day **NS: Strategy Summary** messages are to follow. Other messages can be interspersed with the summaries.

3.13 Trade Correction Messages

3.13.1 XF: Futures Trade Correction (78 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange I. D.		1	A	Exchange on which the trade occurred Q = Montreal
S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year
	Expiry Day	2	N	Future expiry day
Volume		8	N	Total number of contracts traded Refer to Indicator Code
Trade Price		7	N	Price at which the transaction took place
Trade Price Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Price and Fraction Indicator Code
Net Change Sign +/-		1	X	For net change field (sign)
Net Change		7	N	Net change = last trade price – previous settlement price
Net Change Fraction Indicator		1	X	Defines number of decimal places or fraction positions Refer to Price and Fraction Indicator Code
Price Indicator Marker		1	X	Identifies the type of transaction Refer to Price Indicator Markers
Trade Number		8	X	Unique Trade Number for this instrument

3.14 Other Messages

3.14.1 GR: Group Status (40 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal
Root Symbol	6	X	Root of the instrument group
Group Status	1	A	Group status of the trading instrument Refer to Status Markers

This message will be sent when a group of trading instruments enters a new status.

3.14.2 GS: Group Status (Strategies) (36 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal
Group Instrument	2	X	Group of the instrument
Group Status	1	A	Group status of the trading instrument Refer to Status Markers

This message will be sent when a strategy group of trading instruments enters a new status. All strategies have a predetermined group that can be found in the [JS: Strategy Instrument Keys](#) message.

3.14.3 KF: Future Deliverables (min 78 / max 686 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3:
Exchange ID		1	A	Identifies the exchange Q = Montreal
S y m b o l	Root Symbol	6	A	Future base symbol
	Symbol Month	1	A	Future symbol month code Refer to Month Codes
	Symbol Year	2	N	Two last digits of the future symbol year

	Expiry Day	2	N	Future expiry day
	Number of Bonds	2	N	Total number of contracts traded
Up to 20 times	Maturity Date	8	A	Bond expiration
	Coupon	7	N	Bond Coupon rate
	Coupon Fraction Indicator	1	A	Coupon rate decimal indicator Refer to Fraction Indicator Code
	Outstanding Bond Value	8	N	Outstanding amount (CAN \$ million)
	Conversion Factor	7	N	Conversion factor
	Conversion Factor Fraction Indicator	1	A	Conversion factor decimal indicator Refer to Fraction Indicator Code

3.14.4 L: Bulletins (114 Bytes)

Bulletins will be sent throughout the trading day. More than one message will be used if the bulletin is longer than 79 characters. The continuation character '0' indicates that the bulletin continues to the next record.

When a Trading instrument has been halted by the Bourse de Montréal Inc., a Bulletin Message explaining the reason for the halt will be transmitted. When the trading instrument is reinstated, another bulletin message explaining the news that accompanied the reinstatement will be transmitted.

All records that make up a particular bulletin will be sent out together. No other message will be interspersed among the records that make up a complete bulletin.

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3 :
Reserved	1		Reserved for future use
Bulletin Type	1	X	1 = Regular text bulletin, refer to Bulletin Type 1 - Regular Text Bulletin 2 = Special text bulletin, refer to Bulletin Type 2 - Special Text Bulletin

Bulletin Type 1 – Regular Text Bulletin

Field Name	L	T	Definition / Validation Rules
Bulletin Contents	79	X	Bulletin in textual form. Left justified, blank fill
Continue Marker	1	N	0 = Bulletin continues in next record

			1 = Bulletin ended
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Bulletin Type 2 – Special Text Bulletin

This bulletin type will be used for bulletins relating to a specific trading instrument.

Field Name	L	T	Definition / Validation Rules
Symbol	30	X	Symbol
Bulletin Contents	49	X	Bulletin in textual form. Left justified, blank fill
Continue Marker	1	N	0 = Bulletin continues in next record 1 = Bulletin ended

Any continuation records will also contain the symbol as the first 30 bytes of the bulletin field.

3.14.5 S: End of Sales (39 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3 :
Reserved	1		Reserved for future use
Time	6	N	Time at which the message is transmitted, HHMMSS

The **S: End of Sales** message is sent when there is no more trading activity to be transmitted. This occurs after the closing of the market.

3.14.6 TT: Tick Table (Min 103 Bytes / Max 567 Bytes)

Field Name		L	T	Definition / Validation Rules
Message Header		32		Refer to Table 3 :
Exchange ID		1	A	Identifies the exchange Q = Montreal
Tick Table Name		50	X	Tick Table name Refer to Tick Table
Tick Table Short Name		2	X	Tick Table short name Refer to Tick Table
Nb. Entries		2	N	Indicates the number of entries in the table, 1 – 30
Entries (1-30 blocks)	Min Price	7	N	Starting price for this tick entry

	Min Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code
	Tick Price	7	N	Used for this instrument if the price is higher than Min Price
	Tick Price Fraction Indicator	1	X	Defines number of decimal places or fraction positions Refer to Fraction Indicator Code

The **TT: Tick Table** message is used to send the Tick Table information via HSVF.

3.14.7 U: End of Transmission (39 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Exchange ID	1	A	Identifies the exchange Q = Montreal
Time	6	N	Time at which the message is transmitted, HHMMSS

This message will be sent daily to indicate that the day's transmission is complete. No HSVF messages will be transmitted afterwards. Transmission will resume the following day.

3.14.8 V: Circuit Assurance (38 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
Time	6	N	Time at which the message is transmitted, HHMMSS

This message is sent out if no messages are sent by SAM for more than one minute after the broadcast has started (i.e., at the termination of the Test Loop message). This will be an assurance that the line is up.

This message will continue to be sent until the **U: End of Transmission** message is sent. The Circuit Assurance message will repeat the sequence number of the previous record transmitted (except if it is a re-transmit message), i.e. it will not increase the sequence number

3.14.9 SD: Start of day (41 Bytes)

Field Name	L	T	Definition / Validation Rules
Message Header	32		Refer to Table 3:
ExchangeID	1	A	Identifies the exchange Q = Montreal
Business Date	8	N	Current Business date, YYYYMMDD

This message is sent at the beginning of the day to provide the Business Date.

Section 4 Fields Description

4.1 Price and Fraction Indicator Code

The **Price** field is a seven-character numeric field.

Note: The exception to the above is for MarketOnOpen (MOO) orders, where the Price field contains '0000OUV' with a Fraction Indicator Code of '0'. '0000OUV' stands for 'Opening Price' as calculated by the trading engine during the pre-opening phase.

The delineation of the whole number portion of the price and the decimal/fractional portion of the price will be defined by the Fraction Indicator (FI) Code. Furthermore, the FI indicates the manner in which the price is displayed visually. This implies that all zero fractions may be sent in order to maintain consistency in the visual alignment of the implied decimal places. The all zero fraction is replaced by spaces for visual display.

No truncation of price data is permitted by this specification, except for high-order zeros for products that trade in fractions of 1/10,000,000 or smaller. Should such a truncation be necessary, then it is implicit from the FI, which is 7, 8, or 9.

All fractions are expressed as fractions or in decimals as defined by the price fraction rules of the particular product (section [Tick Table](#)). The **Fraction Indicator Code** is one alphanumeric character as follows:

Fraction	Code	Fraction	Code	Fraction	Code	Fraction	Code
1/1	0	-1/1	A			Empty	" "
1/10	1	-1/10	B	10	Z		
1/100	2	-1/100	C	100	Y		
1/1,000	3	-1/1,000	D	1,000	X		
1/10,000	4	-1/10,000	E	10,000	W		
1/100,000	5	-1/100,000	F	100,000	V		
1/1,000,000	6	-1/1,000,000	G	1,000,000	U		
1/10,000,000	7						
1/100,000,000	8						
1/1,000,000,000	9						

4.2 Options Markers

First Letter (Currency or Type of Market)

Marker	Description
B	Trading in British Pound
C	Trading in Canadian Dollar
D	Danish Krone
E	Trading in Swiss Franc
F	Trading in Euro
N	Norwegian Krone
S	Swedish Krona
U	Trading in US Dollar
Y	Trading in Japanese Yen
X	British Pence
2nd letter (Type of Options)	
Blank	Regular Options

4.3 Status Markers

Status		Used in	
Marker	Description	Group Messages	Instrument Messages
Y	Pre-opening	X	X
O	Opening	X	X
T	Continuous Trading	X	X
F	Forbidden	X	X
E	Intervention before Opening	X	X
H	Halted Trading	X	X
R	Reserved (goes into a state as pre-opening where orders can be sent, modified, or canceled)		X
S	Suspended (goes into a state as pre-opening where orders can be sent, modified, or cancelled)		X
A	Surveillance intervention (Consultation phase)	X	X

C	End-of-Day Inquiries	X	X
BLANK	If not used		

4.4 Price Indicator Markers

Price Indicator	
Marker	Description
A	As-of-Trade
B	Block Trade
BLANK	Actual transaction took place
D	Crossed
E	EFP reporting
G	Contingent Trade
H	Riskless Basis Cross
I	Implied Trade
J	Delta Trade
K	Committed Block
L	Late Trade
M	For future use
N	For future use
P	Strategy reporting
Q	For future use
R	EFR reporting
S	Reference Price (volume field zero filled)
t	Trade Correction
T	Committed
U	Basis on Close
V	Price/Volume adjustment
Y	For future use
Z	For future use

4.5 Indicator Code

This code is used for Bid/Ask Size, Volume and Open Interest.

Marker	Description (the size of the bid/ask field is in ...)	
C	100	(Hundreds)
D	1,000	(Thousands)
E	10,000	(Ten-Thousands)
F	100,000	(Hundred-Thousands)
G	1,000,000	(Millions)
H	10,000,000	(Ten-Millions)
I	100,000,000	(Hundred-Millions)
J	1,000,000,000	(Billions)
Data	Message sent	Participant should display...
Bid size of 124,872	Size field will indicate '1248C'	124,800
Volume of 8,457,188	Volume will indicate '8457188'	8,457,188
Volume of 258,487,797	Volume will indicate '2584877C'	258,487,700
Open Interest of 544,871	Size field will indicate '544871'	544,871
Open Interest of 17,458,795	Size field will indicate '174587C'	17,458,700

4.6 Currency Codes

Currency	
Marker	Description
USD	US \$
CAD	Canadian \$
Blank	Not provided

4.7 Line Definitions

Line Name	Instrument Types	Depth
O1	Equity Options Exchange Traded fund (ETF) Options	Best limit

	Index Options Currency Options	
O2	Equity Options Exchange Traded fund (ETF) Options Index Options Currency Options	5 levels
F1	Interest Rate Options Futures Basis on Close	Best limit
F2	Interest Rate Options Futures Basis on Close	5 levels

4.8 Error Definitions

ErrorCode	ErrorMsg
0001	Unknown Message Type
0002	Invalid Message Length
0003	Invalid Characters
0100	Invalid Protocol Version
0101	Already Logged In
0102	Not Logged In
0103	Invalid Line Name
0104	Maximum Number of Retransmission Request Reached
0105	Invalid Sequence Number
0106	Continuous Feed Request in Progress
0107	Cannot Process Legacy Mode Command. TCP Mode Already in Use
0108	Cannot Process TCP Mode Command. Legacy Mode Already in Use

4.9 Protocol Version

Value	Label
D7	D7

4.10 Market Flow Indicators

First Letter	Type of Instrument	Second Letter	Type of Underlying
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F	Futures	U	Short Term Interest Rate
P	Options on Futures	X	Index
O	Options	E	Equities
S	Basis on Close	C	Currency
U	Strategies on Options on Futures	B	Bonds
V	Strategies on Futures	A	ETF
W	Strategies on Options	I	Inter Sectors
Y	Strategies on Inter Commodities	Z	Other
Note that other indicators could be determined later			

4.11 Month Codes

Options and Strategies on Options

Call Options and Strategies		
A – January	E – May	I – September
B – February	F – June	J – October
C – March	G – July	K – November
D – April	H – August	L – December
Put Options		
M – January	Q – May	U – September
N – February	R – June	V – October
O – March	S – July	W – November
P – April	T – August	X – December

Futures / Futures Options / Strategies

F – January	N – July
G – February	Q – August
H – March	U – September
J – April	V – October
K – May	X – November
M – June	Z – December

4.12 Tick Table

This table contains the Tick Table message related information.

TickTableName	ShortName	MinPrice	TickPrice
HalfTick	HT	0.005	0.005
Penny	PE	0.01	0.01
Penny to Dime	2P	0.01	0.01
		0.1	0.05
Nickel	NK	0.05	0.05
Multi Milli	2M	0.001	0.001
		0.01	0.005
Milli	MM	0.001	0.001
Penny to 3 Dollars	3P	0.01	0.01
		3	0.05
5 Dollars	D5	5	5
Dollar	D1	1	1
Dime	DI	0.1	0.1

Note : Order prices will follow the tick table. Trade prices can be more granular than the Tick Table and have up to 4 decimals.

4.13 Strategy Code

This table contains the definition of every available strategy code. The strategy code describes how Leg Instruments relate to each other.

Code	Description
20	option 2 call combination
21	option call put combination
22	option 2 put combination
30	option 3 call combination
33	option 3 put combination
3X	option 3 leg call put combination
40	option 4 call combination
44	option 4 put combination

4X	option 4 leg call put combination
B1	1yr bundle (can be white, red, or green pack)
B2	2yr bundle
B3	3yr bundle
BB	curve butterfly
BC	call butterfly
BP	put butterfly
BS	box spread
CC	call condor
CF	covered future
CP	put condor
CR	credit risk trade
CV	covered options
DC	diagonal call spread
DP	diagonal put spread
FB	future butterfly
FC	future condor
FR	curve trade
FS	roll spread
FT	curve roll
GT	guts
HC	ratio call spread 1x3
HP	ratio put spread 1x3
IB	iron butterfly
IC	iron condor
LC	call ladder
LP	put ladder
MC	call spread vs higher strike put
MP	put spread vs lower strike call
NC	call spread vs lower strike put
NE	custom strategy
NP	put spread vs higher strike call
SC	call spread
SG	strangle
SP	put spread
ST	straddle
TC	ratio call spread 1x2
TP	ratio put spread 1x2
TS	diagonal straddle spread

YL	synthetic
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4.14 Strategy Type

Strategy type describes the method of pricing used for the strategy

"C"	Classic
"S"	Strip
"V"	Covered Regular
R"	Covered reference Fixed

Section 5 Processing Strategies

5.1 Messages to Use

- JS (Strategy Instrument Keys)
- J/JF/JB (Leg Instrument Keys for Options, Futures, and Future Options)

5.2 Process

The following procedure outlines the steps to perform when creating leg & strategy descriptions.

Steps	Description
1	For each J/JF/JB message received, create and maintain a table of the 'leg' couples, ' HSVF Symbol - GroupInstrument/Instrument '.
2	In the JS message, for each leg received: The Leg GroupInstrument/Instrument is linked to the HSVF Symbol from each J/JF/JB message received. Generate the Leg Description by joining: • Leg Ratio Sign • Leg Ratio • Leg GroupInstrument/Instrument Remove all the insignificant " 0 " and spaces. JS messages can be broadcasted anytime during the trading day.
3	Generate the Strategy Description by joining each Leg Description in their respective order in the JS message, and remove all insignificant spaces.

The examples on the following pages are colour-coded as per the following legend:

- Message Type
- HSVF Symbol
- Strategy Symbol
- Instrument External Code
- # of legs
- Leg GroupInstrument/Instrument
- Leg Ratio + Leg Ratio Fraction Indicator
- Leg Price + Leg Price Fraction Indicator

Example 1: Futures vs. Futures detail

0000000134JF20211201190000001000QBAX
 Z1718171218009999000001009989030098090300000053FUBX1162
 BAXZ17 0000250000010003CADBAX
 C

0000000138JF20211201190000001000QBAX
 M1818180618009999000001009979030097990300000103FUBX1762
 BAXM18 0000250000010003CADBAX
 C

0000000142JF20211201190000001000QBAX
 Z1817181217009999000001009966030097860300000103FUBX1D62
 BAXZ18 0000250000010003CADBAX
 C

0000006763JS20211201190000001000QBAX+Z7-2M8+Z8
 17Z18009999000001000050030000500D000000053VUXG50N4
 BAX+Z7-2M8+Z8 YBFC03
 BX1162000100000000
 BX17620002A0000000
 BX1D62000100000000

Msg	Hsvf Symbol	Ext. Instrument Code	Group Instrument	Instrument
JF	BAX Z1718	BAXZ17	BX	1162
JF	BAX M1818	BAXM18	BX	1762
JF	BAX Z1817	BAXZ18	BX	1D62

Msg	Strategy Symbol	Ext. inst. Code	#Legs	Leg Descr. 1	Leg Descr. 2	Leg Descr. 3
JS	BAX+Z7-2M8+Z8	BAX+Z7-2M8+Z8	03	BX1162 00010 0000000	BX1762 0002A 0000000	BX1D62 00010 0000000

Example 2: Options vs. Options detail

```

0000041332J 20211201190000001000QRUS R002700031716
CAD0999990000010999900300000103TT=OP AOER3M900
RUS 170616P27.00 C RUS
0000010000010003CADC

0000041316J 20211201190000001000QRUS R002500031716
CAD0999990000010999900300000103TT=OP AOER3M500
RUS 170616P25.00 C RUS
0000010000010003CADC

0000059697JS20211201190000001000QRUS_UDS_ESMGH6
17F16099999000001099990020999900C00000103WEESMGH6
RUS_UDS_ESMGH6 YSP02
R3M900000100000000
R3M5000001A0000000

```

Msg	Hsvf Symbol	Ext. Instrument Code	Group Instrument	Instrument
J	RUS R 002700031716	RUS 170616P27.00	R3	M900
J	RUS R 002500031716	RUS 170616P25.00	R3	M500

Msg	Strategy Symbol	Ext. Instrument Code	#Legs	Leg Descr. 1	Leg Descr. 2
JS	RUS_UDS_ESMGH6	RUS_UDS_ES MGH6	02	R3M900 00010 0000000	R3M500 0001A 0000000

Example 3: Futures Options and Futures Options (2 legs)

0000003244JB20211201190000001000QOBX H1819C00992503
 180319CAD0099990000010000350300000013TT=OP
 PUOX3GWC OBXH18C9925 0000000102500002CADCBAX
 H18

0000003284JB20211201190000001000QOBX H1819P00992503
 180319CAD0099990000010000650300000503TT=OP
 PUOX3HWC OBXH18P9925 0000000102500002CADCBAX
 H18

0000006765JS20211201190000001000QH8STDL92
 18H19009999000001099990020999900C00000053UUXG50P4H8STDL92
 YSD02
 OX3GWC000100000000
 OX3HWC000100000000

Msg	Hsvf Symbol	Ext. Instrument Code	Group Instrument	Instrument
JB	OBX H1819C00992503	OBXH18C9925	OX	3GWC
JB	OBX H1819P00992503	OBXH18P9925	OX	3HWC

Msg	Strategy Symbol	Ext. Instrument Code	#Legs	Leg Descr. 1	Leg Descr. 2
JS	H8STDL92	H8STDL92	02	OX3GWC 00010 0000000	OX3HWC 00010 0000000

Example 4: Futures Options and Futures (2 legs Covered)

0000003284JB20211201190000001000QOBX H1819P00992503
 180319CAD0099990000010000650300000503TT=OP
 PUOX3HWC OBXH18P9925 0000000102500002CADCBAX
 H18

0000000134JF20211201190000001000QBAX
 H1818180318009999000001009989030098090300000053FUBX1162
 BAXH18 0000250000010003CADBAX
 C

0000006765JS20211201190000001000QCV_XG50P4
 18H19009999000001099990020999900C00000053UUXG50P4H8STD92
 YCVV2021120116300002
 OX3HWC04251000000
 BX11620001000099622

Msg	Hsvf Symbol	Ext. Instrument Code	Group Instrument	Instrument
JB	OBX H1819P00992503	OBXH18P9925	OX	3HWC
JF	BAX H1818	BAXH18	BX	1162

Msg	Strategy Symbol	Ext. Instrument Code	#Legs	Leg Descr. 1	Leg Descr. 2
JS	CV_XG50P4	CV_XG50P4	02	OX3HWC 04251 0000000	BX1162 00010 00099622



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